

Cash Crunch Autopilot: Agentic AI for Small Business Cash Safety on Razorpay

An Agentic AI Finance Guardian for Bharat's Small Businesses — Powered by Razorpay's Payment, Banking, Lending and Payroll Data

An autonomous AI agent that looks at a merchant's incoming sales, bank balance, loan EMI dates, and payroll dates together — something no single Razorpay product does today — and predicts if the merchant will run short of cash before payday. It doesn't just warn the merchant, it suggests a clear, low-cost fix and asks for approval before doing anything with real money.

Theme: AI Finance Controller — Building trust in automated financial decisions

Target Area: Razorpay Payment Gateway, RazorpayX, Razorpay Capital, Razorpay Payroll

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The Problem Statement

Why Good Businesses Miss Payroll — Even When They're Doing Fine

[Link for video](#)

[Pitch](#)



The Small Business Owner's Reality

A small business owner selling online gets paid through Razorpay, keeps money in a Razorpay current account, is repaying a Razorpay Capital loan, and pays staff through Razorpay Payroll. All four of these are Razorpay products — but the owner has no single place that shows if all four numbers will add up on payday.

The Gap in Current Systems

Today, Payment Gateway, RazorpayX, Capital, and Payroll all work as separate products. None of them talk to each other. A business can have healthy sales and still run out of cash on the 1st of the month because a loan EMI and a payroll date happen to fall on the same day. No one is watching all four together.

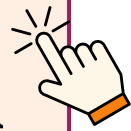
The Painful Surprise

The business owner usually finds out they are short on cash only when the payroll payment actually fails or bounces. This is one of the most stressful moments for a small business - employees not getting paid on time. By the time it happens, it is too late to fix cheaply.

We chose this problem because **Razorpay's biggest strength** is that it already sits on all four pieces of a business's money life — **sales, bank balance, loans, and payroll**. Our vision is not to build the flashiest tool, but the most useful one: an AI system that quietly watches the numbers and warns a business before a crisis, not after.

Cash Crunch Autopilot: From Silos to Signals

[Link for
Documentation](#)



What's Broken Today, What We Built, and How It Actually Works

• Existing Solution

Cash flow tools today (banking apps, spreadsheets, even Razorpay's own dashboards) show one number at a time - just the bank balance, or just the loan due date, or just sales. The business owner has to manually stitch these together in their head.

• The Gap

None of these tools combine sales pattern + bank balance + loan due date + payroll due date for the same business, in real time, to predict a shortfall before it happens.

• Our Solution

Cash Crunch Autopilot uses Agentic AI to continuously watch all four signals together for each merchant. It doesn't just show numbers - it predicts the exact day a shortfall will happen, explains why, suggests a specific low-cost fix, and asks the merchant to approve before touching any money.

1.

Gather (Data Collection)

Automated Retrieval

The agent pulls the merchant's recent sales trend (from Payment Gateway), current bank balance (from RazorpayX), upcoming loan EMI (from Capital), and next payroll amount and date (from Payroll)

2.

Predict (Forecast Engine)

Strict Mathematical Calculation

A simple day-by-day cash projection model calculates the expected bank balance for each of the next 14 days, based on the sales trend and the two upcoming outgoing payments.

3.

Reason (LLM Orchestration)

Adding Real-World Context

The AI agent looks at the numbers and adds real business context - for example, telling the difference between "a slow week" and "an actual emergency" - and decides which fix makes the most sense for this merchant.

4.

Decide (Compliant Lock)

The Final Audit

The final recommendation is checked against fixed rules (maximum advance amount, verified product features only) so the agent never suggests something Razorpay cannot actually do.

5.

Act (Autonomous Execution)

Zero Human Intervention

The agent shows the merchant the shortfall, the fix, and the exact cost. Nothing happens until the merchant taps "Approve." The agent never moves money on its own.

Technical Architecture

[Link for More
Diagram](#)



Tech Stack

Frontend:



React.js + TailwindCSS
(Merchant dashboard,
Admin dashboard)

Backend:



Node.js + Express.js
(orchestration layer,
REST APIs)

ML Layer:



Python
(simple forecasting
model + risk
scoring)

Agentic Framework:



LangChain.js
(task routing,
reasoning chain)

Infra:



Vercel + Render
(Free Tier Support)

Banking:

**Razorpay Payments ,
x payroll , xbusiness
, capital**

LLM (Open Source):



Open-source model
(for explanation and
recommendation
reasoning)

Database:



MongoDB
(merchant
transaction and
forecast data store)

Lending & Payroll (Simulated)

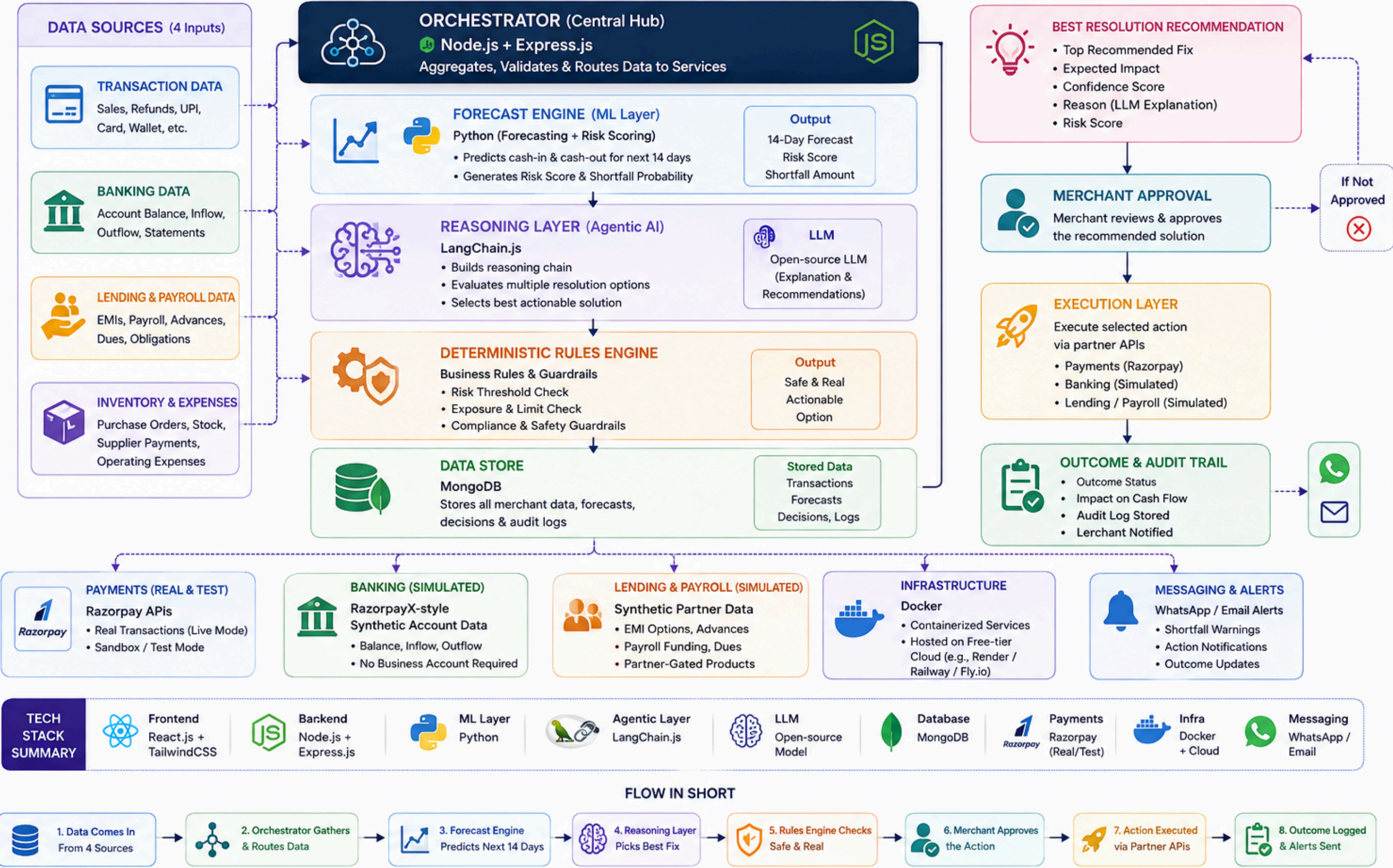
**Realistic synthetic
data (partner-gated
products)**

Multilingual Support:



google Cloud
translate API

Agentic Cash-Flow Guardrail System Architecture



Technical Excellence & AI + ML Architecture



Machine Learning Layer

Aspect	Implementation
Method	Day-by-day cash balance projection (rolling average of sales minus known outgoings)
Why this method	Simple, fast, fully explainable — no black box needed for a money decision
Input Features	Daily sales (60-day history), current bank balance, loan EMI date & amount, payroll date & amount
Training/Testing Data	50 synthetic merchants, with 10–15 merchants deliberately set up to run short of cash
Output	Predicted shortfall day, shortfall amount, confidence level
Explainability	Every prediction shows the exact numbers used to reach it
Decision Time	Under one second per merchant

Agentic AI Layer



Why This Architecture?

Forecast Model	Agentic AI
Predicts the shortfall	Explains the shortfall in plain language
Calculates exact numbers	Understands business context
Fast and repeatable	Compares trade-offs between fixes
Rule-checked	Talks to the merchant conversationally

Expected Model Performance

Metric	Target
Shortfall Detection Accuracy	85%+
False Alarm Rate	Under 15%
Average Prediction Lead Time	5–7 days before shortfall
Metric	Target

Impact on Merchants & Razorpay

For Small Businesses:



Early Warning

Know about a cash shortfall 5–7 days before it happens, not on the day it fails.



Cheaper Fixes

Get the lowest-cost option instead of panicking and taking an expensive loan.



No Missed Payroll

Employees get paid on time, protecting trust and morale.

For Razorpay (indirect revenue chain):



More Product Usage

Every suggested fix (like an instant settlement) uses a real Razorpay paid feature.



Stronger Merchant Retention

Merchants who feel protected stay longer on the platform.



Cross-Sell Opportunity

Merchants without a Capital loan or Payroll account may sign up for one after seeing the benefit.

[Link for
Documentation](#)



Key Features

Smarter Cash Flow. Less Stress. Built for Every Merchant.



1. No Manual Bookkeeping Needed

Uses data Razorpay already has.



2. Predicts Shortfalls up to 2 Weeks Ahead

Helps you plan before it becomes critical.



3. Explainable Reasoning for Every Alert

See why the alert is raised and what's driving it.



4. Merchant Approval Required Before Any Action

You stay in control of every step.



5. Shows Exact Cost of Every Suggested Fix

No surprises. Just clear numbers.



6. Weekly Automatic Re-forecast

Always stays updated with your latest data.



7. Works Even for Businesses with No Formal Accounting

No spreadsheets or complex records needed.



8. Simple Dashboard

Red flag on merchants about to run short.



9. Privacy-First

Only reads data merchant already shares with Razorpay.

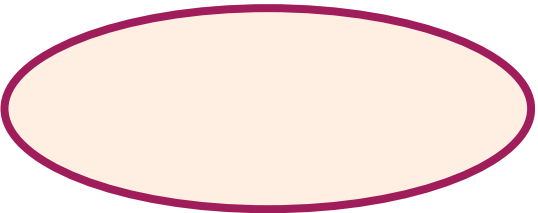


10. Built to Scale

To lakhs of merchants using the same logic.

Theme Alignment

- **"AI Finance Controller"** - Directly matches the brief's own example: a forward cash forecaster.
- **"Power of Agentic AI"** - The agent doesn't just calculate a number; it reasons about the best fix, checks it against real rules, and only acts after human approval -like a digital finance assistant.



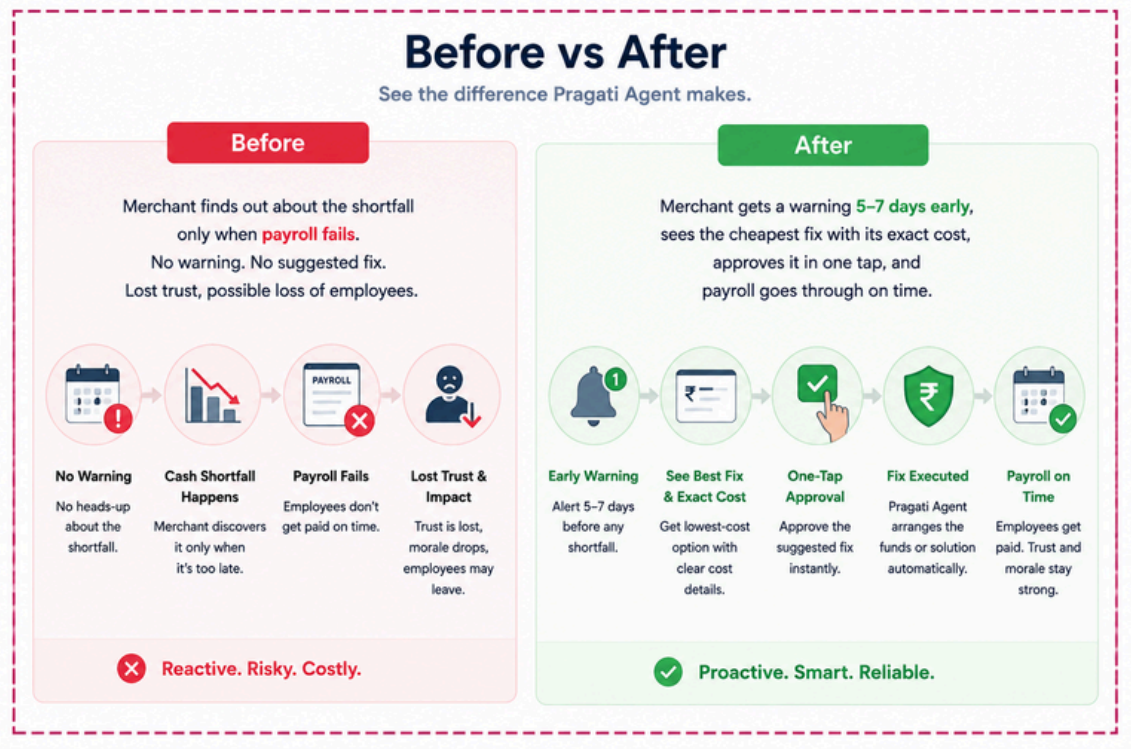
The Business Case: Cost Feasibility



Cost to Implement

Component	Technology	Estimated Cost
Forecast Engine	Python (simple math model)	Very Low
LLM Reasoning	Open-source / pay-per-token API	Low-Medium
Backend APIs	Node.js + Express	Low
Database	MongoDB	Pay-as-you-scale
Alerts	WhatsApp/Email API	Low (per message)
Infra	Free-tier cloud hosting	Low

- Missed payroll is a well-known, painful problem for small businesses everywhere.
- Cash flow tools exist for single bank accounts (like Float, QuickBooks) but none combine sales + banking + loan + payroll for the same business the way Razorpay's own data can.
- This is a problem only Razorpay is positioned to solve well, because it already holds all four pieces of data.



Before vs After

- **Before:** Merchant finds out about the shortfall only when payroll fails. No warning. No suggested fix. Lost trust, possible loss of employees.
- **After:** Merchant gets a warning 5–7 days early, sees the cheapest fix with its exact cost, approves it in one tap, and payroll goes through on time.

Feasibility

- **Uses data Razorpay already collects** - no new data pipeline needed.
- **Low technical risk** - the forecast is simple arithmetic, not a complex model.
- **Fits directly into the existing merchant dashboard.**

Scalability

- **The same forecast logic works for 50 merchants or 5 lakh merchants.**
- **Lightweight** - no heavy AI compute needed for the core prediction.
- **LLM is only used for the explanation step, keeping running costs low.**